

Date

2013

Square Star Pattern:-

* for n=5:-

(i) rows = n (outer loop)

column = n (inner loop)

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
  
```

* Logic:-

i (Row)	star
1	→ 5
2	→ 5
3	→ 5
4	→ 5
5	→ 5

* Pseudo codes:-

- Repeat n times
- Repeat n times
- print *
- Move to the next line

* Code:-

```

int n=5
for(int i=1; i<=n; i++){
    for(int j=1; j<=n; j++){
        System.out.print("* ");
    }
    System.out.println();
}
  
```

→ print *

→ move to next line.

Output

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
  
```

=== Code Execution Successful ===

⇒ Hollow Square Star Pattern :-

* for $n=5$, $i=1 \rightarrow$ $j=1$ $j=n$

```

* * * * *
*       *
*       *
*       *
* * * * *
  
```

$i=n \rightarrow$

* Pseudo code :-

```

for i=1 to n
  for j=1 to n
    if i==1 || i==n || j==1 || j==n
      print "*"
    else
      print " "
  
```

print new line

int $n=5$;

* Code :-

```

for (int i=1; i<=n; i++){
  for (int j=1; j<=n; j++){
    if (i==1 || i==n || j==1 || j==n){
      System.out.print("*");
    } else {
      System.out.print(" ");
    }
  }
  System.out.println();
}
  
```

* Logic :-

Row (i)	Star
first row	all stars
last row	all stars
first column	all stars
last column	all stars
Remaining positions	spaces

Output

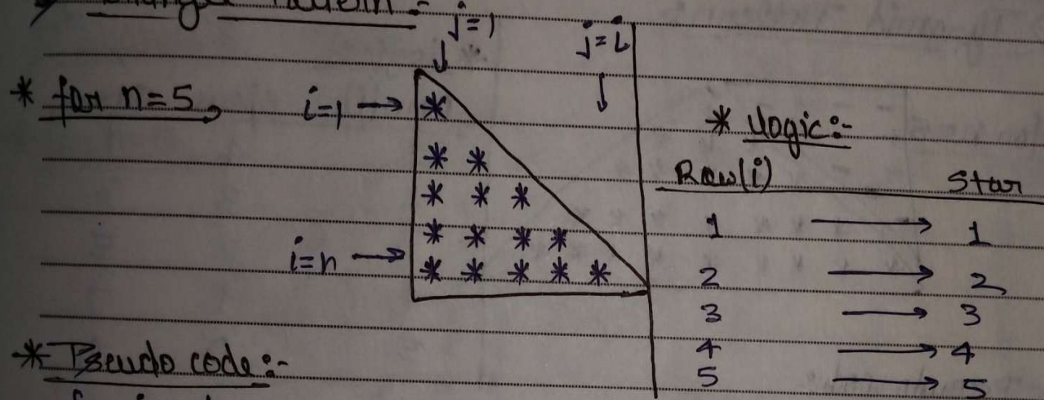
```

*****
*       *
*       *
*       *
*       *
*****
  
```

=== Code Execution Successful ===

Date

2013

3) Triangle Pattern:-* Pseudo code:-

```

for i=1 to n
  for j=1 to i
    print "*"
  print new line
  
```

* Code:-

```

int n=5;
for(int i=1; i<=n; i++){
  for(int j=1; j<=i; j++){
    System.out.print(" * ");
  }
  System.out.println();
}
  
```

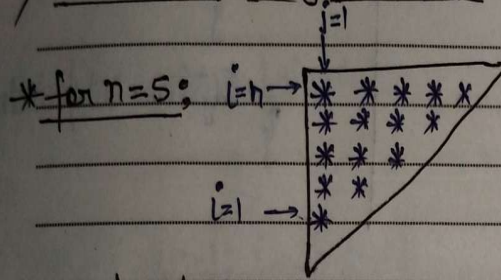
Output

```

*
* *
* * *
* * * *
* * * * *
  
```

=== Code Execution Successful ===

4) Inverted Triangle Pattern:-



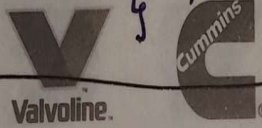
* Pseudo code:-

```
for i=1 to n
  for j=1 to (n-i+1)
    print "*"
  print new line
```

* Code:- `int n=5;`

```
for (int i=1; i<=n; i++){
  for (int j=1; j<=n-i+1; j++){
    System.out.print("* ");
  }
  System.out.println();
}
```

`System.out.println();`



Output

```
* * * * *
* * * *
* * *
* *
*
```

=== Code Execution Successful ===

Date.....

➤ Pyramid Pattern:-

* for n=5;

```

  *
 * *
* * *
* * * *
* * * * *
  
```

* Logic:-

Row(i)	spaces	Star
1	4	1
2	3	3
3	2	5
4	1	7
5	0	9

* Pseudo code:-

```

for i=1 to n
  for space=1 to n-i
    print space
  for star=1 to 2*i-1
    print "*"
  print new line
  
```

* Code:-

```

int n=5;
for(int i=1; i<=n; i++){
  for(int j=1; j<=n-i; j++){
    System.out.print(" ");
  }
  for(int k=1; k<=(2*i-1); k++){
    System.out.print("*");
  }
  System.out.println();
}
  
```

Output

```

  *
 * *
* * *
* * * *
* * * * *
  
```

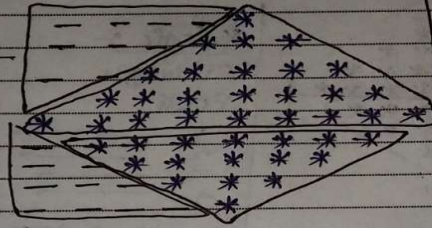
=== Code Execution Successful ===

Date.....

2013

6) Diamond Pattern:-

* for n=5,

* Logic:-① Pyramid

Row(i)	Spaces	Star
1	4	1
2	3	3
3	2	5
4	1	7
5	0	9

② Inverted Pyramid↓
Inverse version of Pyramid* Code:-

```

int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=n-i; j++){
        System.out.print(" ");
    }
    for(int k=1; k<=2*i-1; k++){
        System.out.print("*");
    }
    System.out.println();
}

for(int i=n-1; i>=1; i--){
    for(int j=1; j<=n-i; j++){
        System.out.print(" ");
    }
    for(int k=1; k<=2*i-1; k++){
        System.out.print("*");
    }
    System.out.println();
}

```

* Pseudo code:-① Upper half

```

for i=1 to n
    print spaces(n-i)
    print stars(2*i-1)
    new line

```

② Lower half

```

for i=n-1 down to 1
    print spaces(n-i)
    print stars(2*i-1)
    new line

```

Output

```

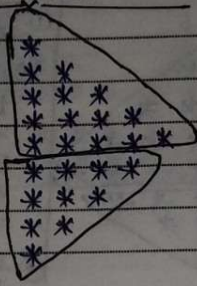
*
***
*****
*****
*****
*****
***
*

```

=== Code Execution Successful ===

⑦ Half diamond Pattern:

* for n=5:



* Logic :-

① Triangle

Row(i)

Star

1	→	1
2	→	2
3	→	3
4	→	4
5	→	5

② Inverted Triangle

↳ It reverse of triangle

* code:-

```
int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=i; j++){
        System.out.print("* ");
    }
    System.out.println();
}

for(int i=n-1; i>1; i--){
    for(int j=1; j<=i; j++){
        System.out.print("* ");
    }
    System.out.println();
}
```

* Pseudo code:-

1) upper half;

for i=1 to n

print i stars
new line

2) lower half;

for i=n-1 down to 1

print i stars
new line

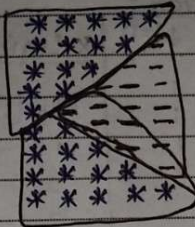
Output

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

=== Code Execution Successful ===

8) Half inverted diamond pattern:

* for n=5:



* Logic:

① Inverted Triangle

+

② Triangle

* Code:-

```
int n=5;
for(int i=n; i>=1; i--){
    for(int j=1; j<=i; j++){
        System.out.print("*");
    }
    System.out.println();
}

for(int i=2; i<=n; i++){
    for(int j=1; j<=i; j++){
        System.out.print("* ");
    }
    System.out.println();
}
```

* Pseudo code:-

① Upper half

for i=n down to 1
print i stars
new line

② Lower half

for i=2 to n
print i stars
new line

Output

```
*****
*****
****
***
**
*
**
***
****
*****
```

=== Code Execution Successful ===

Date: _____

1) Square Number Pattern:

* for n=5:

$i=1 \rightarrow$ 1 1 1 1 1
 $i=2 \rightarrow$ 1 1 1 1 1
 $i=3 \rightarrow$ 1 1 1 1 1
 $i=4 \rightarrow$ 1 1 1 1 1
 $i=5 \rightarrow$ 1 1 1 1 1

* Logic:

Row(i) print 1

1 $\rightarrow$ 5
 2 $\rightarrow$ 5
 3 $\rightarrow$ 5
 4 $\rightarrow$ 5
 5 $\rightarrow$ 5

* Code:- `int n=5;`
`for(int i=1; i<=n; i++){`
`for(int j=1; j<=n; j++){`
`System.out.print("1 ");`
`}`
`System.out.println();`
`}`

* Pseudo code :-

for i=1 to n
 for j=1 to n
 print 1
~~End for~~
 print new line

Output

```
11111
11111
11111
11111
11111
```

=== Code Execution Successful ===

2) Incrementing Square Pattern:-

* for n=5,

```

1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5

```

* Logic :-

Row(i) print

```

1 → 1 to 5
2 → "
3 → "
4 → "
5 → "

```

* Code:- int n=5;

```

for(int i=1; i<=n; i++){
    for(int j=1; j<=n; j++){
        System.out.print(j+"");
    }
    System.out.println();
}

```

* Pseudo code:-

```

for i=1 to n
    for j=1 to n
        print j
    print new line

```



Output

```

12345
12345
12345
12345
12345

```

=== Code Execution Successful ===

Date.....

2013

3) Incrementing Right Triangle Pattern:-* for n=5,

```

1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

* Logic:-

Row(i)	print
1	→ 1
2	→ 1 to 2
3	→ 1 to 3
4	→ 1 to 4
5	→ 1 to 5

* Code:-

```

int n=5;
for(int i=1; i<=n; i++){
    for(int j=1; j<=i; j++){
        System.out.print(j+" ");
    }
    System.out.println();
}

```

* Pseudo code:-

```

For i=1 to n
    for j=1 to i
        print j
    print new line

```

Output

```

1
12
123
1234
12345

```

=== Code Execution Successful ===

4) Inverted diamond Pattern:

* for n=5,

```

3 3 3 3 3
4 4 4 4
5 5 5
6 6
7

```

* Logic:-

- Spaces = i $\Rightarrow$ increase
- Number = 3 + i $\Rightarrow$ increase
- Count = n - i $\Rightarrow$ decrease

* Code:

```

int n=5;
int num=3

for(int i=0; i<n; i++){
    for(int j=0; j<i; j++){
        System.out.print(" ");
    }

    for(int k=0; k<n-i; k++){
        System.out.print(num + " ");
    }

    num++;
    System.out.println();
}

```

* Pseudo code:-

```

num = 3
For i = 0 to n-1
    print i spaces
    print num (n-i) times
    num++
    print new line

```

Output

```

3 3 3 3 3
4 4 4 4
5 5 5
6 6
7

```

=== Code Execution Successful ===

Date _____

2013

5) Sandwich Pattern:

```

* for
  3 3 3 3 3
  4 4 4
  5
  4 4 4
  3 3 3 3 3

```

* Logic :-

- Number: $3 \rightarrow 4 \rightarrow 5 \rightarrow 4 \rightarrow 3$
- Count: $5 \rightarrow 3 \rightarrow 1 \rightarrow 3 \rightarrow 5$

* Code:

```

for(int i=0; i<3; i++){
  for(int j=0; j<i; j++){
    System.out.print(" ");
  }
  for(int k=0; k<5-(2*i); k++){
    System.out.print((3+i)+" ");
  }
  System.out.println();
}

for(int i=1; i>=0; i--){
  for(int j=0; j<i; j++){
    System.out.print(" ");
  }
  for(int k=0; k<5-(2*i); k++){
    System.out.print((3+i)+" ");
  }
  System.out.println();
}

```

* Pseudocode:

```

For i=0 to 2
  print i spaces
  print (3+i) (5-2*i) times
  print new line

For i=1 down to 0
  print i spaces
  print (3+i) (5-2*i) times
  print new line

```

Output

```

3 3 3 3 3
4 4 4
5
4 4 4
3 3 3 3 3

```

=== Code Execution Successful ===

Date _____

2013

6) Incrementing Diamond Pattern:-

```

*
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3

```

* Logic:-

Number: 3 → 4 → 5 → 6 → 7

$$\downarrow$$
 6 → 5 → 4 → 3

Count = No. of elements in row

* Code:-

```

for(int i=1; i<=5; i++){
    int num=i+2;
    for(int j=1; j<=i; j++){
        System.out.print(num+" ");
    }
    System.out.println();
}

for(int i=4; i>=1; i--){
    int num=i+2;
    for(int j=1; j<=i; j++){
        System.out.print(num+" ");
    }
    System.out.println();
}

```

* Pseudo code:-

For i=1 to 5
 print (i+2) i times
 print new lines

for i=4 down to 1
 print (i+2) i times
 print new line

Output

```

3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3

```

=== Code Execution Successful ===